HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed **C/C++** modules for real-time operating system (RTOS) components, contributing to a **15% reduction** in system boot time for a new chip architecture.
- Authored **Python** scripts to automate the generation of test cases and validation checks for embedded firmware, increasing test coverage by **22%**.
- Integrated and debugged **ARM and RISC-V assembly language** code segments within a larger C++ codebase for performance-critical operations.
- Collaborated with cross-functional teams to identify and resolve architectural and design issues in multi-core computer architectures, improving overall system stability.
- Implemented **embedded software security** best practices, including secure bootloader sequences and encrypted communication protocols.
- Contributed to the setup and maintenance of **CI/CD pipelines** using **Jenkins** and **Github Actions** for automated builds and testing of embedded firmware.
- Participated in on-call rotations, performing root cause analysis for critical production issues and implementing timely fixes within a **Linux** environment.

SKILLS

Embedded Systems – RTOS, C, C++, Assembly language, RISC-V, ARM, Multi-Core Computer Architecture, Embedded software debugging, JTAG, Trace32, OpenOCD, Segger J-Link, LLDB, Secure Software development practices.

Automation & Scripting – Python, Shell scripting.

Development Tools – Jenkins, Github Actions, Gerrit, Gitlab, Docker.

Operating Systems – Linux, RTOS.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch).

PROJECTS

RTOS Kernel Enhancement for Low-Power IoT Devices

- Developed **C** modules for a real-time operating system kernel, focusing on optimizing power consumption for battery-operated IoT devices, achieving a **10% improvement** in battery life.
- Implemented interrupt handling routines and task scheduling algorithms to ensure deterministic execution under high load conditions, maintaining **p95 latency below 5ms**.
- Authored detailed design documents and unit tests, contributing to a **20% reduction** in post-release defects for the kernel subsystem.

Automated Firmware Validation Suite

- Created a **Python**-based framework to automate the validation of embedded firmware features, integrating with hardware-in-the-loop test benches.
- Designed and implemented test scripts that covered over **150 distinct firmware functionalities**, significantly speeding up the regression testing cycle.
- Utilized **Git** for version control and collaborated with QA engineers to refine test case coverage and reporting mechanisms.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)